S : Introduction to Fractions

For the tutor Fractions mean equal parts of a whole. Always begin with drawings (circles, rectangles, food examples). Do not rush into rules. Make sure the student understands that the bottom number tells how many equal parts the whole is divided into.

Section A Understanding Unit Fractions

For the tutor: A unit fraction has 1 on top. It represents one equal part of a whole.

1. Shade $\frac{1}{2}$ of the shape (e.g. circle, rectangle, etc.).
2. Shade $\frac{1}{3}$ of the shape.
3. Shade $\frac{1}{4}$ of the shape.
4. Circle the unit fractions:
   $\frac{1}{5}$ $\frac{2}{3}$ $\frac{1}{8}$ $\frac{3}{4}$
5. What does the bottom number in $\frac{1}{6}$ tell us?

Section B Equal Parts or Not?

For the tutor: Fractions only work if parts are equal.

1. A pizza is cut into 4 equal slices. Is one slice $\frac{1}{4}$ of the pizza? Yes No
2. A cake is cut into 3 pieces, but one piece is much bigger. Can we call one piece $\frac{1}{3}$? Yes No
3. A rectangle is divided into 5 equal parts. What fraction is one part? _____
4. If a chocolate bar is divided into 8 equal pieces, what fraction is one piece? _____

Section C Fractions on the Number Line

For the tutor: Fractions are numbers. They live between 0 and 1 on the number line. Make sure they can use a number line to understand how much fractions divide. Draw a number line on your table and ask the questions shown below.

1. Mark $\frac{1}{2}$ on the number line below:

   0 ——————————— 1

2. Mark $\frac{1}{4}$ on the number line below:

   0 ——————————— 1

3. Mark $\frac{1}{3}$ on the number line below:

   0 ——————————— 1

Section D Comparing Unit Fractions

For the tutor: When the top number is 1, the bigger the bottom number, the smaller the piece. You can use shapes to help understand the intuition behind this.

1. Which is larger?

   $\frac{1}{2}$ or $\frac{1}{4}$

2. Which is larger?

   $\frac{1}{3}$ or $\frac{1}{6}$

3. Which is smaller?

   $\frac{1}{5}$ or $\frac{1}{8}$

4. True or False:

   $\frac{1}{7}$ is larger than $\frac{1}{3}$

5. True or False:

   $\frac{1}{1}$ is smaller than $\frac{1}{2}$

Section E Word Problems

Read and answer

1. Ko ate $\frac{1}{4}$ of a cake. Was the cake cut into 4 equal parts or 1 equal part?

2. Ama drank $\frac{1}{2}$ of her juice. If the bottle was full, how many equal parts was it divided into?

3. A rope is cut into 5 equal pieces. What fraction of the rope is one piece?

Section F Explain in Your Own Words

For the tutor: Listen carefully to how the student explains.

1. What is a unit fraction?

2. Why is $\frac{1}{8}$ smaller than $\frac{1}{4}$?